

xVox

Total Services Framework

SOHO Broadband

Wholesale Peering

Enterprise VPN

Callback Services

Calling Cards



Managed VoIP Solutions

SOHO Broadband

A component of the xVox Total Services Framework

User Manual

Windows or Mac

Internet Explorer v5

January 15, 2005
DOCS-200451501



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Getting Help

The xVox platform is a sophisticated and complex system that requires detailed knowledge to configure. It can be easy for you to modify parameters so that the system does not behave in the manner that you expect.

Telic's support staff is available 24x7 via email, instant message and phone to provide assistance. To reach them, use any of the following methods:

- **Browse the support site.** For support information, such as release notes, guides and templates, visit the Telic Technical Support Web site at:

<https://support.telic.net>

- **Send an Email.** Send comments or problems with Telic products to **support@telic.net**.
- **Talk to a support staff.** Use the following numbers to contact the NOC Support Engineers:

+1 212 400-8181 ext. 1104 [US] or

+63 2 747-3460 ext. 1104 [Philippines]

(Phone support is only available to extended support customers)

The onus is on you, the system administrator, to configure your xVox services in the way you want. Telic's technical support is not a substitute for training or commissioning. If you need training on the product, please contact your sales representative to make arrangements. Use Telic's technical support to solve problems that you have tried diligently to solve by yourself.

Purchasing Additional Services

If you need information on purchasing Telic products, contact a Telic sales representative through the following:

- Email: **sales@telic.net**
- Phone: **+1 212 400-8181 ext. 1010 [US]**

For additional information about Telic Communications, Inc. and Telic products, visit the Web site at:

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I. Overview

I.A. Introduction to the xVox Total Services Framework

The guiding principle behind our technology is to provide a single real time platform for deploying and managing next generation services across multiple networks (circuit, packet, wireless and cable), and wherever possible, extending the best features of each to all.

Telic executes this vision with the xVox platform, a multi-service software system that offers a single point from which to develop, deploy, manage and bill for network services across heterogeneous networks. Consisting of multiple modularized components that may be deployed individually to fit the needs of each customer, xVox is a high performance, fault-tolerant software platform for developing and deploying revenue-generating network services.

The precept behind xVox is to separate the business and service logic from the underlying switching elements in a network, and to facilitate easy integration with existing systems. This enables faster and more flexible network services delivery because new components can be added, or an existing component can be modified, without the need to re-engineer all other parts of the system.

The xVox platform achieves these goals by adhering to industry-wide standards for telecom integration, including OSA/PARLAY and OSS/J. Being OSA/Parlay compliant means that xVox is designed to work together with both current and next generation networks because applications are created and delivered with network independent APIs. OSS/J allows the xVox platform to exchange information with other OSS/BSS platforms in real time, thereby reducing the need to maintain multiple silos of replicated data.

I.B. Introduction to the Real Time Billing Environment (RTBE)

The Real Time Billing Environment, or RTBE, is the centerpiece of the xVox platform. It is a complete and customizable IP billing solution that enables service providers to control all their customer management and billing functions across a variety of business models. The RTBE is ideally suited to provide access control to any IP-based service, including VoIP calling services, Prepaid Internet access or Prepaid Internet gaming.

The RTBE has been engineered from the ground up with flexibility in mind. The RTBE engine employs a modular client/server architecture and a user-definable rating structure. This allows the service provider to adapt its usage to virtually any imaginable type of access control and/or billing scenario.

I.C. Concepts

The following are general concepts that form the basics of the RTBE:

I.C.1 Rating

At the heart of the RTBE is a flexible real-time rating mechanism that allows service providers to readily accommodate new services. Each service type has a separate base rate table, which defines the units and quantity of the transaction that is being measured. Service type base rates may be defined to apply during specific times or days. Once defined, these service types may then be modified by any number of user-configurable charge types. Charge types are used to add incremental charges to a transaction such as a tax expressed as a percentage of a per-minute tariff, or a per-session charge incurred per use. Charge types may also be used to reduce a base rate, and therefore allow selective discounting based on call/session duration.

I.C.2 Routing

The RTBE is also capable of routing VoIP calls between various VoIP devices. This is a singular feature that differentiates the RTBE system from virtually every other VoIP billing system available.

The RTBE protocol was designed to draw route information at the same time that it performs authentication of the originating party and rating of the call. The route selection criteria is based on a combination of the route's price, quality and reliability. Accordingly, the RTBE allows for destinations to be defined to include any combination of user-configurable information associated with a VoIP gateway.

Routing rules may be defined to factor in conditions such as least-cost preference, time-of-day or desired level of service. The RTBE uses two metrics to gauge the quality of a particular IP route: Quality of Service (QoS) and Reliability of Service (RoS). QoS is a user-configurable parameter that allows the service provider to assign a rank to a particular route based on known characteristics. On the other hand, RoS is a dynamic tool that calculates the reliability of a given VoIP route based on the success ratio of prior connections. The RTBE routing capabilities also include support for black lists and location-specific dialing rules.

I.D. Features

The RTBE offers the following features:

I.D.1. Multi-User level Account Management

Data in the RTBE is arranged and presented in multiple levels wherein lower level users can create and manage their accounts without assistance or intervention by the service provider. At the same time, the service provider is able to monitor and control lower level users through configurable service-level boundaries and easily generated reports.

I.D.2. Web-Based Administration

All the components of the RTBE can be configured, monitored and controlled remotely using a web browser.

I.D.3. Report Generator

The RTBE provides users at all levels with real-time access to appropriate data. Powered with a report generator, the RTBE is able to display reports based on account, billing and usage information. The data is presented in detailed report formats that can be customized to meet the needs of the service provider. The RTBE also has several Invoicing features that provide safe, correct billing functions.

I.D.4. IVR Integration

Using Telic's Media Server and its Interactive Voice Response (IVR) application, users can be given instant access to their account information over the telephone. Depending on the application, it is possible not only to have account balance information read over the phone, but also to replenish debit accounts and or pay for services using a credit card payment gateway through the Media Server.

I.D.5. Access Control

The RTBE has the ability to control usage on a per 'account' level. Each account corresponds to a single user (users may be individuals, or companies). Each account may have one or more Personal Identification Numbers (PINs). Each PIN has an associated balance (in the case of prepaid services) or credit limit (in the case of postpaid services). Access to the system is denied in real time once the user has depleted the balance or reached the credit limit of their account.

I.D.6. Reliability

The RTBE servers have been designed to gracefully recover from error conditions such that any loss in connectivity with an RTBE client will not result in an unbilled transaction. This allows the system to operate without losing any data and without interrupting operation of the RTBE client (except in cases where the host of the RTBE client itself fails).

I.D.7. Security

The RTBE employs numerous security mechanisms to safeguard the integrity of account information. Among others, all communication between browsers and the web server utilizes SSL to avoid packet sniffers from retrieving account information.

System administration access can also be limited to a single or known group of IP addresses so that unauthorized users are not given the opportunity to test the system.

I.D.8. OSS Integration through XML Webservices

The RTBE has a pre-defined Application Programming Interface (API) set that allows integration with existing Operation Support Systems (OSS) and other back-office environments. For example, with minimal work, an existing order entry application with an XML interface can be used to populate an account table in the RTBE database using our SOAP-XML webservices. From the perspective of the OSS environment, the RTBE is simply another network element in the existing hierarchy. Similarly, the same XML interface can be used by external applications to query the RTBE database for pertinent information. In this way, an existing billing solution can batch import RTBE data in near-real time in order to generate an invoice.

I.E. System Architecture

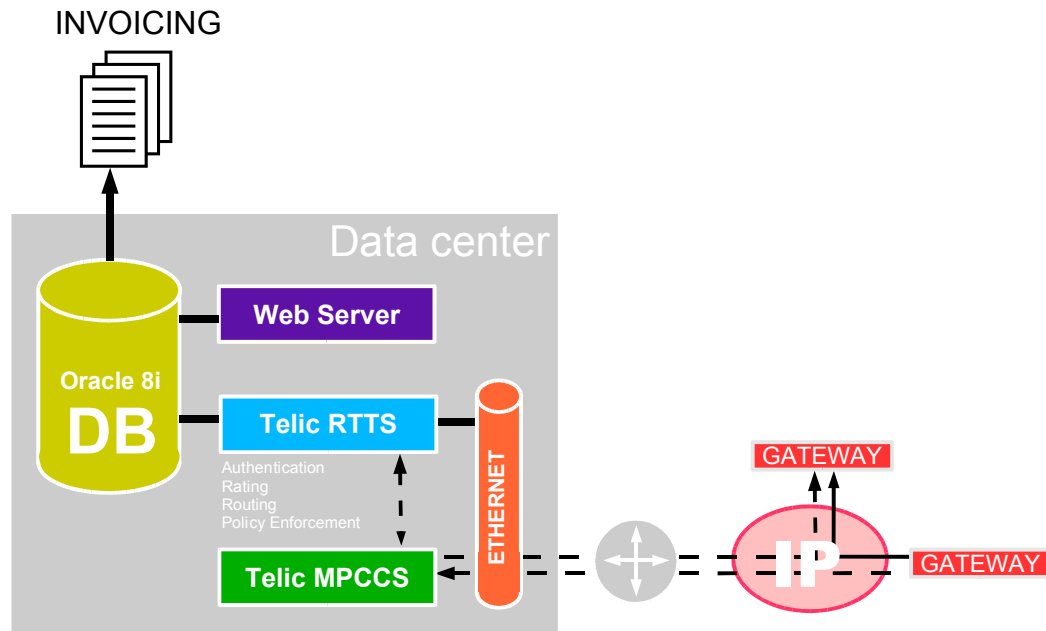


Figure 1. RTBE Architecture

I.E.1. Architecture of Telic's RTBE

The RTBE architecture is comprised of four components (refer to Figure 1):

1. a Real-Time Transaction Server (RTTS) - this is the primary transaction engine;
2. the RTBE Database;
3. the RTBE web servers; and
4. an RTBE client embedded in our Multi-Protocol Call Control Server (MPCCS), also known as a softswitch and Media Server.

Designed to allow interaction by multiple softswitches or media servers with a single data store, the RTBE achieves data integrity and scalability through load balanced RTTS clusters and a redundantly configured database. Data from one or more RTBE clients can be 'balanced' among multiple RTTS in order to optimize performance. With each server performing up to 200 transactions per second, on average, nearly any combination of scale and speed can be reached. Further, in the event of a failure of any single RTTS, the clients will seek the next available RTTS in order to transparently resume operation.

I.E.2. RTBE Hierarchy

The RTBE is built around the following four levels:

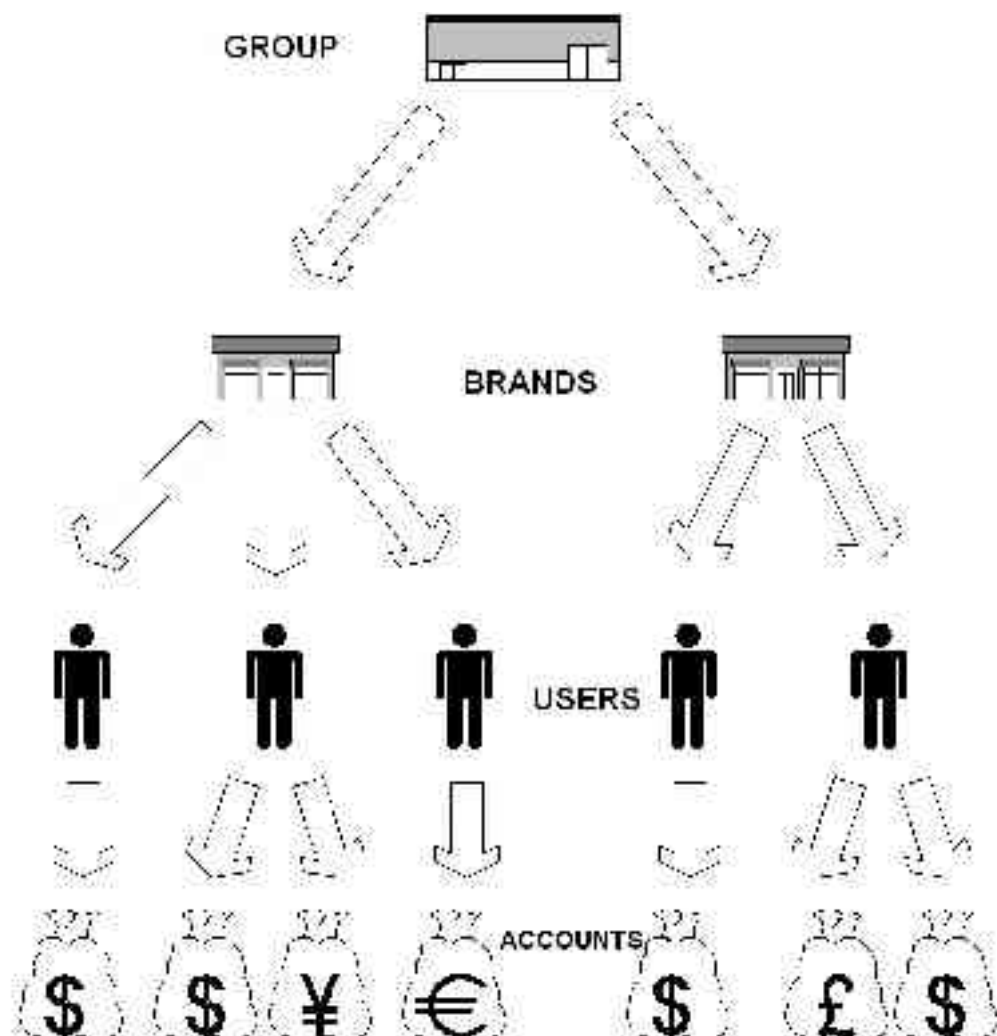


Figure 2. RTBE Hierarchy

1. **Groups:** the highest level in the RTBE. Groups are Telic's clients; they are service providers. Within the system, only Groups have relationships with carriers and can manage routing. Groups create bundles of rates and routes called Group Services. Groups then 'sell' their Group Services to lower-level entities, called Brands.
2. **Brands:** the second highest level in the RTBE. Brands are entities created under the Group in order to partition customers. The Group administrator may create any number of Brands. Each Brand may comprise Products, Base Rates and Charge Types applicable to the demands of a particular customer (or group of customers).
3. **Users:** the end users. A user is any entity that is responsible for one or more PIN Accounts, can be individuals or companies.
4. **PIN Accounts:** Accounts in the RTBE system that track a debit or credit balance and is protected by a secret PIN code. Users may each own any number of Accounts.

I.E.2.a. Groups

As Telic clients, Groups are service providers. Groups maintain relationships with Carriers and can manage routing. They provide their own origination and termination through gateways, or can purchase these services from any third party and incorporate the rates in their offerings.

Groups create Group Services, which are distinct bundles of Products, Routes and Base Rates. The Group Service defines the offering that the Group provides to its Brand(s). Since the Group Service controls the ultimate routing, a Brand must be assigned to at least one Group Service in order to function.

A sample Group Services may include the following information:

Product (usually Destination)	Mexico
Route	Carrier A
Carrier Rate (Exchange Code)	\$.04 per minute
Selling rate	\$.044 per minute

By marking up on the selling rate of the route to the Brands, Groups earn profit.

I.E.2.b. Brands

Brands are entities created by the Group for call tracking purposes. Brands may be distinct entities, such as resellers, that distribute for the Group, or they may simply represent partitions of the Group's customer base.

The Group administrator may create any number of Brands. Each Brand may be tailored to have Products, Base Rates and Charge Types applicable to the demands of a particular group of end customers.

Brands are subscribed to Group Services and create Brand Services . Like a Group Service, a Brand Service is a distinct bundle of Products and Base Rates.

There are two types of Brand Services: wholesale and retail.

In the case of retail Brand Services, routes and rates defined by the Group can be marked-up and made available to the end-users.

Continuing the examples in the previous section on Groups, Brand Services might include:

Group Service	Product (usually Destination)	Mexico
	Route	All calls terminated via Carrier A
	Cost of Route	\$.04
Wholesale Brand Service	Brand Service	Gold Brand
	Rate Charged to Wholesale Customer	\$.044 per minute
Retail Brand Service	Brand Service Name	Premium Calling Card
	Rate Charged to Retail Customer	\$.059 per minute
	Additional Surcharge	\$.025 per call

In addition to being a defined service offering, the Brand Service provides a mechanism for the RTBE to assign transactions to a particular Brand. For example, if Brand A creates a Brand Service named H323:64.95.28.21, all sessions originating from the IP address 64.95.28.21 using the H323 signaling protocol will be assigned to Brand A. Brand A's Products, Rates and Routes will apply to that call.

I.E.2.c. Users

Users are customers of a Brand. A user may be any entity that would typically receive an invoice or statement of use for activity on the system.

I.E.2.d. PIN Accounts

The RTBE uses the terms "PIN" and "Account" interchangeably. As a matter of practice, 'PIN Account' is used. A PIN Account is the container that holds a user's credit or debit balance. A User may have more than one PIN Account. A single PIN Account may not be assigned to more than one User.

I.E.2.e. Additional Attributes

Each level in the RTBE may have one or several of the following attributes:

I.E.2.e.1. Base Rates

Base Rates are the rates that are defined for the Products contained in a Group Service or Brand Service. Each Product must have a Base Rate.

I.E.2.e.2. Products

A Product describes a dial pattern corresponding to a distinct destination. Products that belong to a Group are Group Products; products that belong to a Brand are Brand Products. Products usually comprise of a string of digits representing country code + city code, although it is possible to create products that do not correspond with a valid PSTN number (such as an extension on an IPBX). Unique routes may be defined for each Product using the RTBE routing feature.

I.E.2.e.3. Hosts

A Host may be any device capable of making or receiving IP calls. The RTBE refers to Hosts as either 'originating' or 'terminating'. Hosts include VoIP Gateways, IP Phones, IADs, Media Servers, H323 Gatekeepers, Softswitches, and others. Hosts must be 'registered' as part of a Carrier in order to interact with the RTBE system.

I.E.2.e.4. Carriers

Carriers are logical groupings of Hosts and are used to define routes. Selecting a route for a particular Product is done by specifying a Carrier (as opposed to specifying individual gateways or hosts).

I.E.2.e.5. Agents

Agents can be any entity with a financial interest in the traffic of a given Group Service or Brand Service. Once assigned to a service, agents are able to access the system and monitor usage.

The system is also able to track commissions for Agents on a percentage, per minute, or per-call basis.

II. SOHO Broadband Configuration

II. A. Overview

II.A.1. Introduction to SOHO Broadband

The xVox SOHO Broadband solution converts existing broadband connections (such as xDSL, Fiber, Metro Ethernet or Cable) into an access network for residential and home office telephony service.

With comprehensive back-office features that facilitate all aspects of customer management including prepaid or postpaid billing plans, the xVox SOHO Broadband solution offers any service provider a speedy path to broadband service offerings.

Looking at Figure 3, we can see how a call is made through the SOHO Broadband service.

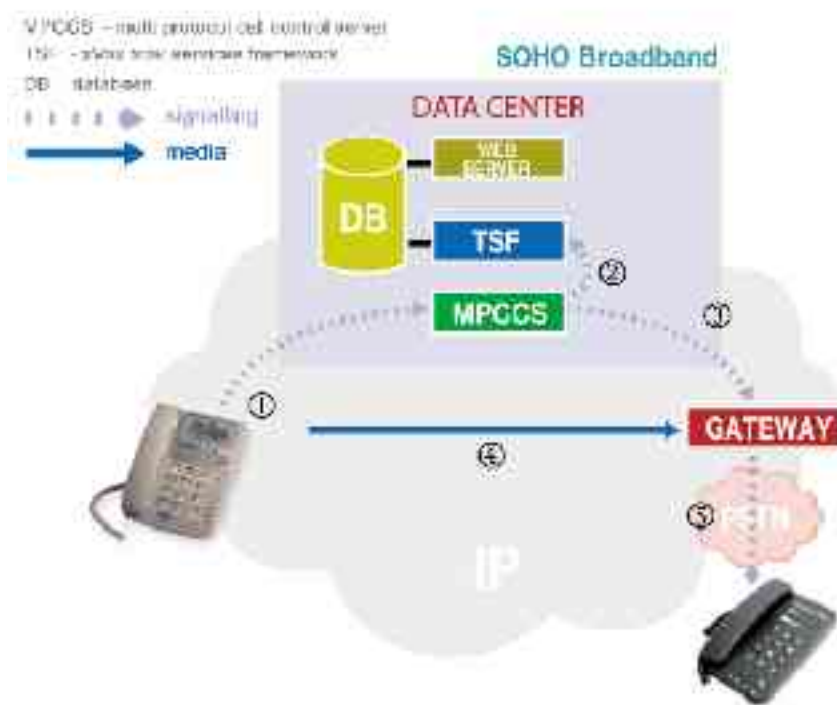


Figure 3. SOHO Broadband

How it works:

1. IP Phone establishes signaling link to MPCCS.
2. MPCCS establishes session with TSF for authentication, rating and routing.
3. MPCCS sets up signaling with terminating gateway.
4. Voice path is established between IP Phone and gateway.
5. Call is connected.

II.A.2. Domain Registration Process

Central to the Retail Softswitching process is the use of a SIP domain for registration by individual SIP Phones. A SIP domain is no different than other internet domains such as 'www.telic.net,' it is simply a means of identifying a specific server on the Internet using a name instead of an IP address. Like any Internet domain, a SIP domain must be a fully qualified domain name (FQDN). In other words, it must be registered with a public DNS server and reachable over the public Internet.



A domain name locates an organization or other entity on the Internet. For example, the domain name 'www.totalbaseball.com' locates an Internet address for "totalbaseball.com" at Internet point 199.0.0.2 and a particular host server named "www". The "com" part of the domain name reflects the purpose of the organization or entity (in this example, "commercial") and is called the top-level domain name. The "totalbaseball" part of the domain name defines the organization or entity and together with the top-level is called the second-level domain name. The second-level domain name maps to and can be thought of as the "readable" version of the Internet address. A third level can be defined to identify a particular host server at the Internet address. In our example, "www" is the name of the server that handles Internet requests. (A second server might be called "www2".) A third level of domain name is not required. For example, the fully-qualified domain name could have been "totalbaseball.com" and the server assumed.

Creating a valid SIP domain requires the use of a DNS server. In many cases, this will be provided by your ISP. If you do not have access to the DNS server of your ISP, there are several commercial DNS services available which will allow you to manage your SIP domains. To find one, simply perform an Internet search for 'managed DNS.' The diagram below shows the domain resolution process.

The steps are (refer to Figure 4):

1. The SIP phone, having been configured to use the SIP domain 'voip.telic.net' will attempt to resolve the domain by querying its configured DNS server;
2. After being configured once, the DNS server has a record (referred to as an 'A record') that resolves the domain 'voip.telic.net' to a telic.net softswitch with the IP address 63.251.28.103;
3. The SIP phone registers with the Telic softswitch and connects to it every time a call is attempted (in this example the Telic softswitch is the 'registrar' and 'outbound proxy').

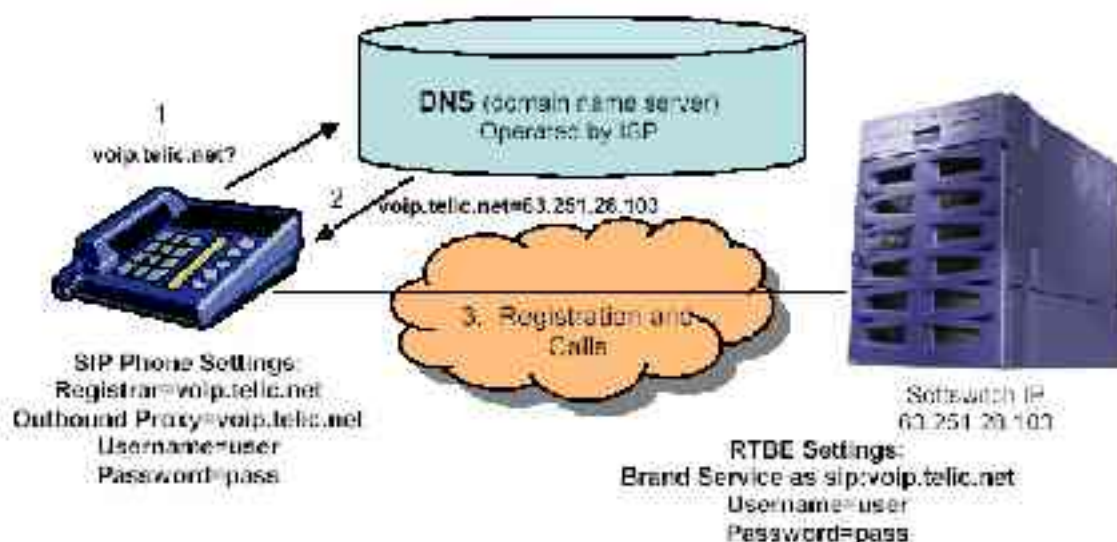


Figure 4. The Domain Registration Process

In the example above, the fact that the domain 'voip.telic.net' points to a Telic softswitch may seem counterintuitive at first. It may help to think of it as a way for the SIP phone to tell the softswitch which Brand Service it belongs to (since the SIP domain is used as the originating host, and the originating host is the basis of a Brand Service name).

This distinction would not be possible if the SIP phone were configured to connect to an IP address instead of a domain name (since many other gateways, using different Brand Services, connect to the IP address of the softswitch).

II.A.3. Identifying End Users

Since many phones will appear as a single RTBE originating host, it is necessary to identify individual phones for proper authorization and billing.

Configuring SIP Phone service requires the creation of RTBE User accounts in order to track the usage of individual phones. As suggested by the diagram in Figure 4, the username and password configured in a SIP Phone must match the username and password established for an individual RTBE User. Thereafter, anytime a call is made from that particular SIP Phone, the calls will be identified in the RTBE as belonging to the matching RTBE User.

II.B. SOHO Broadband Standard Configuration

The following are the steps in configuring the SOHO Broadband service.

II.B.1. Create Group Administrator

To create a Group Administrator, do the following:

- a. Go to **Group>Group Administrators>ADD GROUP ADMINS**.
- b. Click on **ADD GROUPADMINS** to create one or more Administrator accounts for your Group.
- c. Enter the username, password, and desired expiration date for the administrator account to create.
- d. Enter the pertinent data concerning the new administrator using the **profile** link.



The time zone that is selected in an administrator's profile will govern the time zone reflected in all transactions viewed by that administrator in the RTBE Administrator Interface. The RTBE system stores all data in Greenwich Mean Time (GMT). Timestamps for all transactions are adjusted according to the viewer's time zone profile when the RTBE web server renders pages.

1. Set the appropriate time zone in your profile if you are the first Group Administrator. This information is available using the **My Profile** link in the left hand navigation menu. If no time zone is selected in an administrator's profile, then all references to time in the RTBE web user interface use GMT.
2. Assign the new administrator the default Role named **Group Admin**, or, if no default Role is shown, click on the **custom** link from the drop-down menu to establish the desired privileges for the Group Administrator account you are creating.
3. To create a new Role, click on **ADD NEW**. On the **Add Role** Page, enter the Role Description and then check the -box of each privilege that the Role may have access to.
 - a. Save your entries.
- e. The Role you create will be saved in the system so that you may assign it to new Administrators of the same Group in the future.

II.B.2. Create an Originating Carrier

To create an originating carrier, do the following:

- a. Go to **Group>Routed Services>Manage My Carriers**.



Originating Carriers are used in the RTBE to designate the actual operator of VoIP gateways (referred to as 'Hosts' in the RTBE) that will be sending traffic. Once created, Hosts are assigned to a specific Brand. Often, an originating Carrier is an entity that operates a gateway on behalf of a Brand. For example, if Brand Acme uses MCI to originate VoIP traffic (VoIP access), the Carrier would be MCI, although the Group's customer is Brand Acme.

- b. Click on **ADD NEW CARRIER**.
- c. Enter the originating Carrier's name.